

VANGUARD ROBOTICS 2026

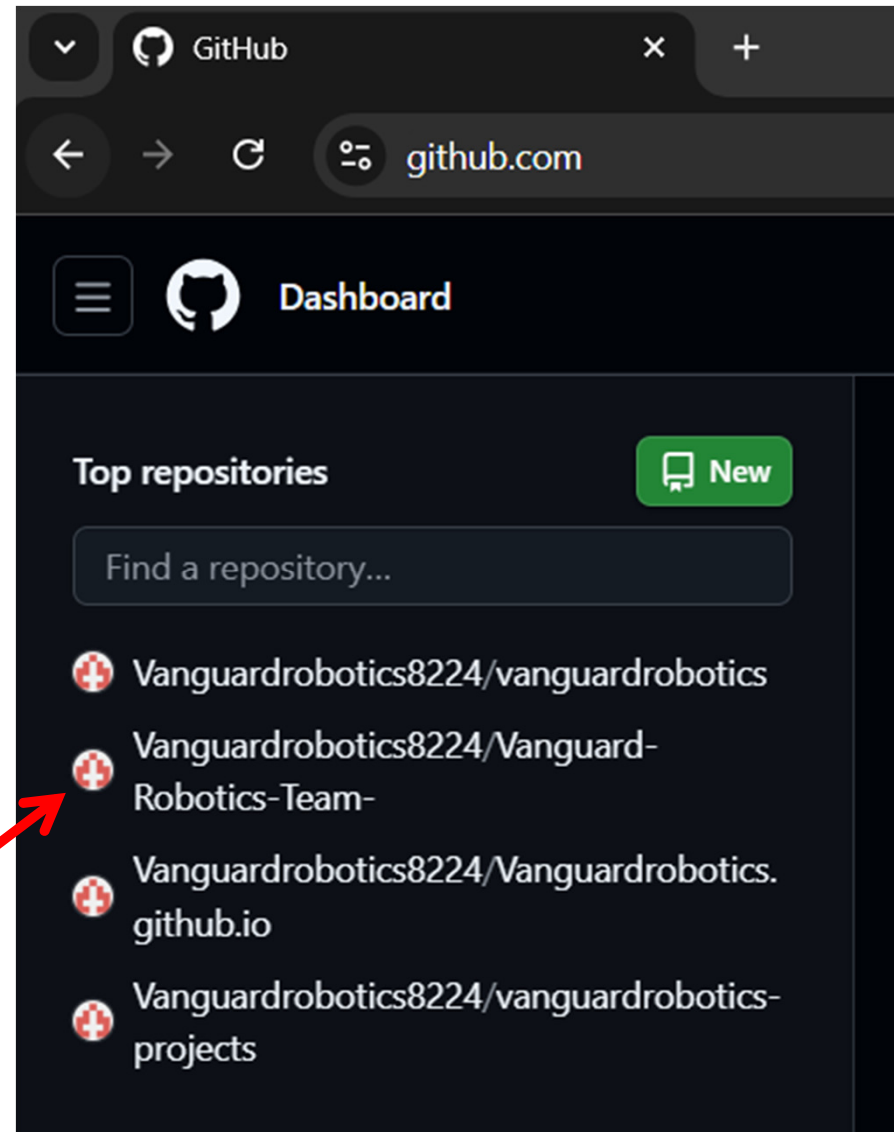
Github / Sourcetree Setup/usage guide
and
File sharing procedures

Sign in to github.com

Note the repositories:

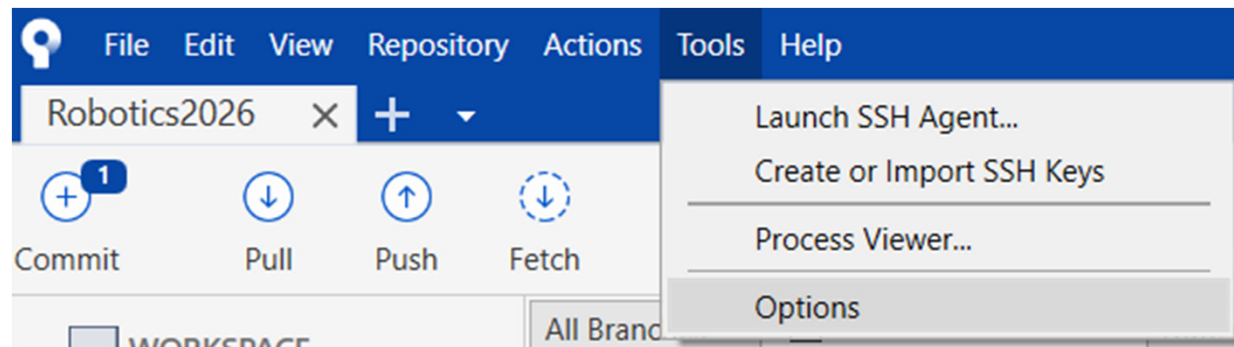
- vanguardrobotics
 - website –do not use
- Vanguardrobotics.github.io
 - website –do not use
- vanguardrobotics-projects
 - sample coding projects
- Vanguard-Robotics-Team
 - main code base

Click on Vanguard-Robotics-Team



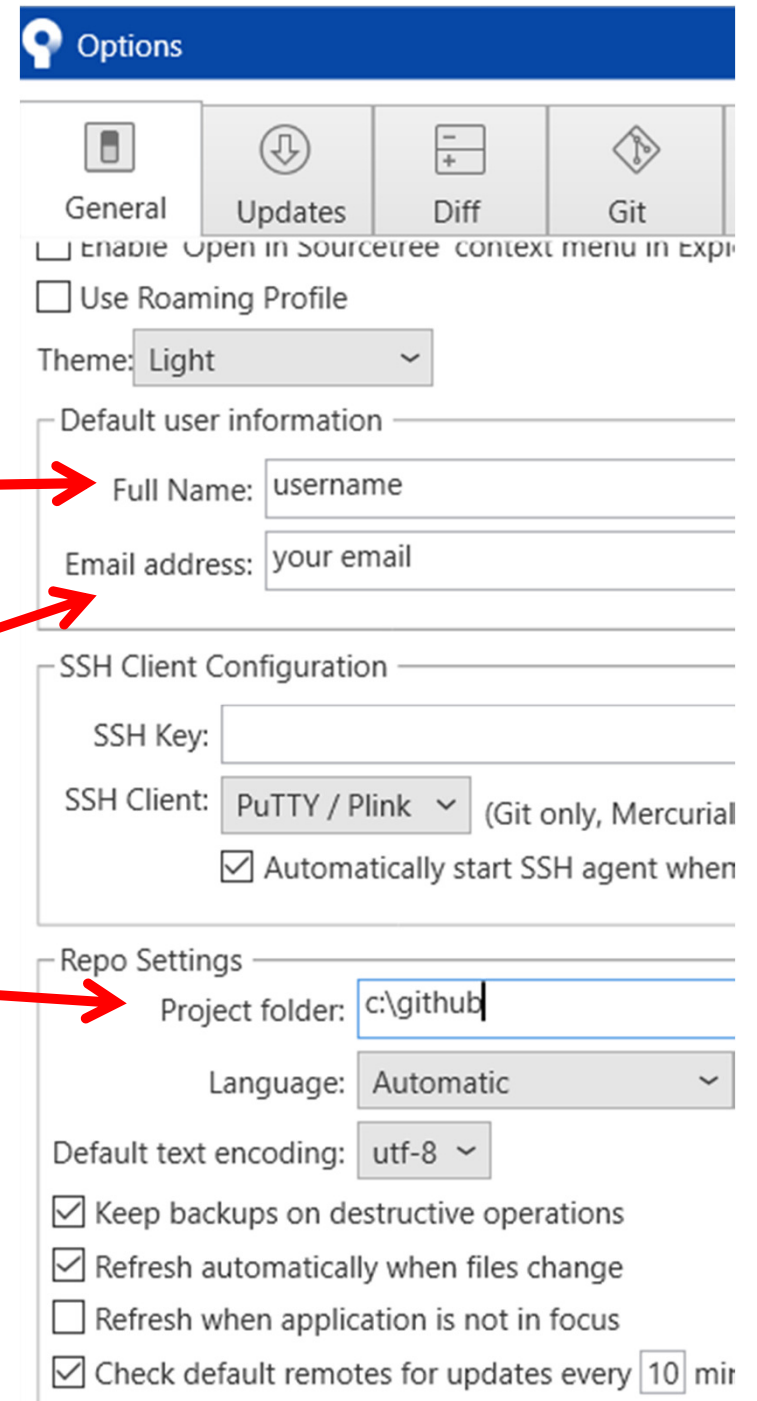
Set up SourceTree

- Open SourceTree. This assumes you signed in to your bitbucket account earlier.
- Go to Tools
 - > Options



Set up SourceTree

- Full Name
 - Your name
- Email address
 - Your email address
- Project Folder
 - C:\github



The screenshot shows the 'Options' dialog box in SourceTree. The 'General' tab is selected. The 'Default user information' section contains fields for 'Full Name' (set to 'username') and 'Email address' (set to 'your email'). The 'SSH Client Configuration' section shows 'SSH Client' set to 'PuTTY / Plink' and 'Automatically start SSH agent when' checked. The 'Repo Settings' section shows 'Project folder' set to 'c:\github', 'Language' set to 'Automatic', and 'Default text encoding' set to 'utf-8'. There are also checkboxes for 'Keep backups on destructive operations', 'Refresh automatically when files change', 'Refresh when application is not in focus', and 'Check default remotes for updates every 10 min'.

Options

General Updates Diff Git

☐ Enable Open in Sourcetree context menu in Explorer

☐ Use Roaming Profile

Theme: Light

Default user information

Full Name: username

Email address: your email

SSH Client Configuration

SSH Key:

SSH Client: PuTTY / Plink (Git only, Mercurial)

☒ Automatically start SSH agent when

Repo Settings

Project folder: c:\github

Language: Automatic

Default text encoding: utf-8

☒ Keep backups on destructive operations

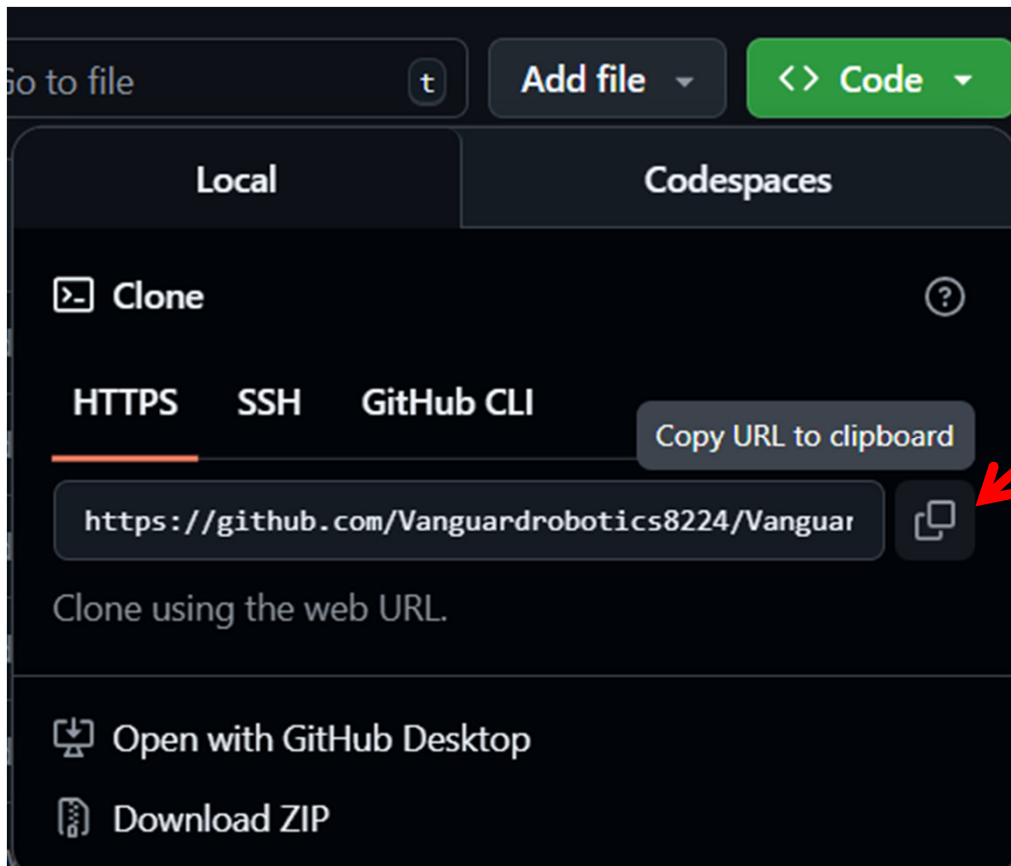
☒ Refresh automatically when files change

☐ Refresh when application is not in focus

☒ Check default remotes for updates every 10 min

Clone a Repository

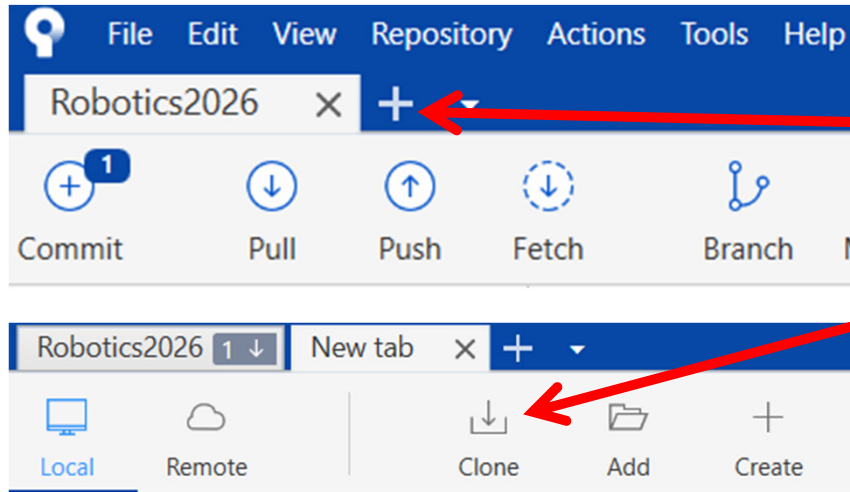
step 1 -github



- Click the green Code button
- Select the copy button
 - This will copy the url for the github repository to your computer's clipboard

Clone a Repository

step 2 -SourceTree



- Click the + button
- Select Clone

Local repositories

Search

Clone a Repository

step 2 -SourceTree

Robotics2026 1 ↓ New tab × + ▾

Local Remote Clone Add Create

Clone

Cloning is even easier if you set up a remote account

Browse

Repository Type: This is a Git repository

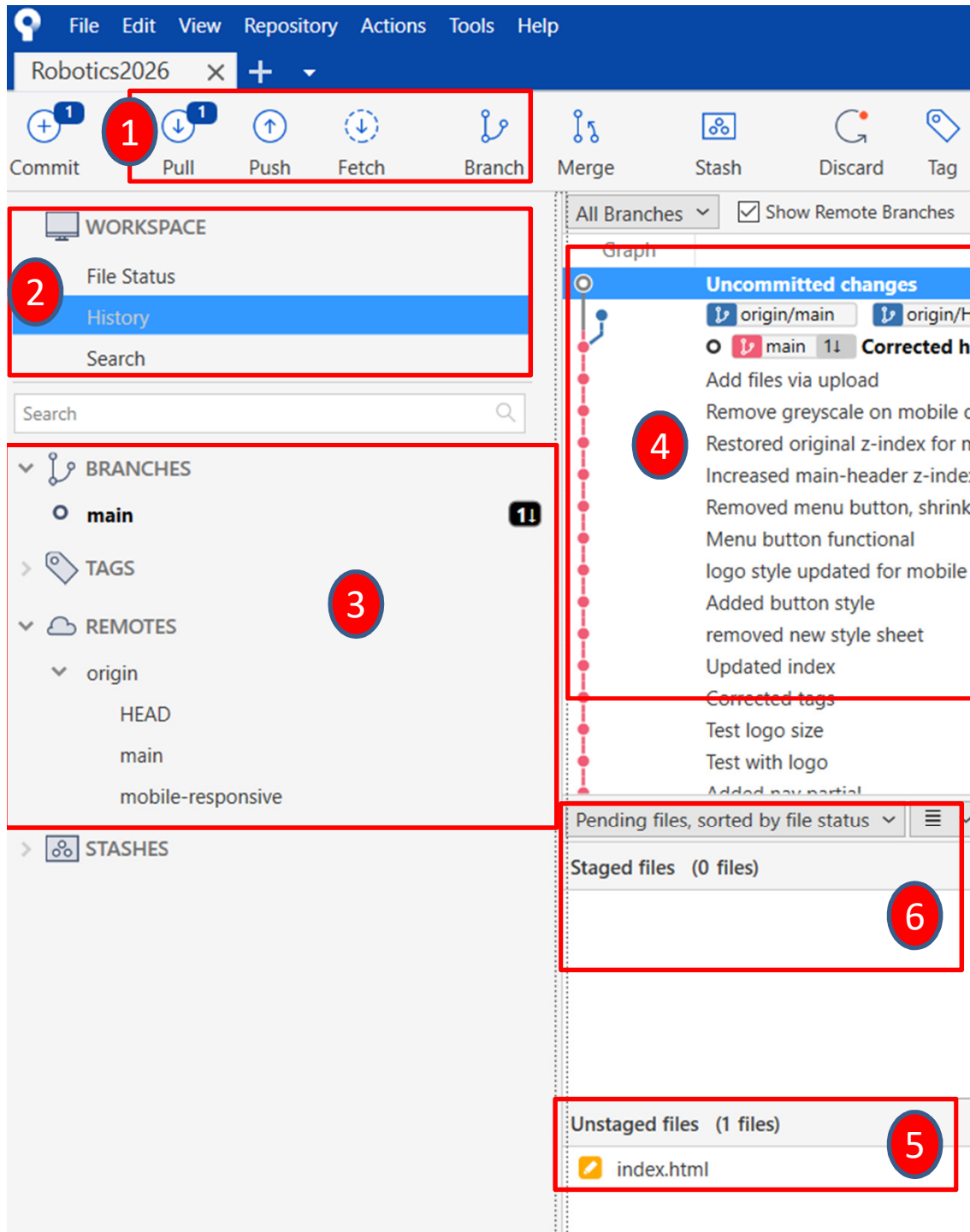
Browse

Local Folder:

> Advanced Options

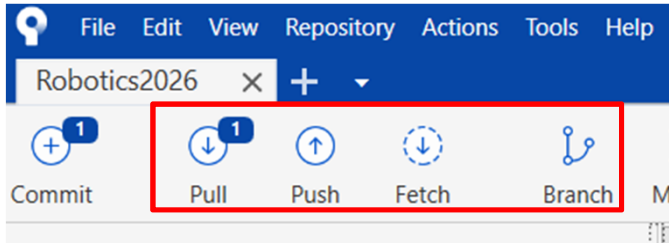
Clone

- Paste the github repository url here
- The other fields should update automatically
- Click Clone



Working with SourceTree

- 1 Buttons used most often
- 2 Different views of window
- 3 Shows all the branches in github or on your laptop
- 4 List of all commits (changes) made
- 5 List of files changed on your laptop
- 6 List of files ready to commit



Working with SourceTree

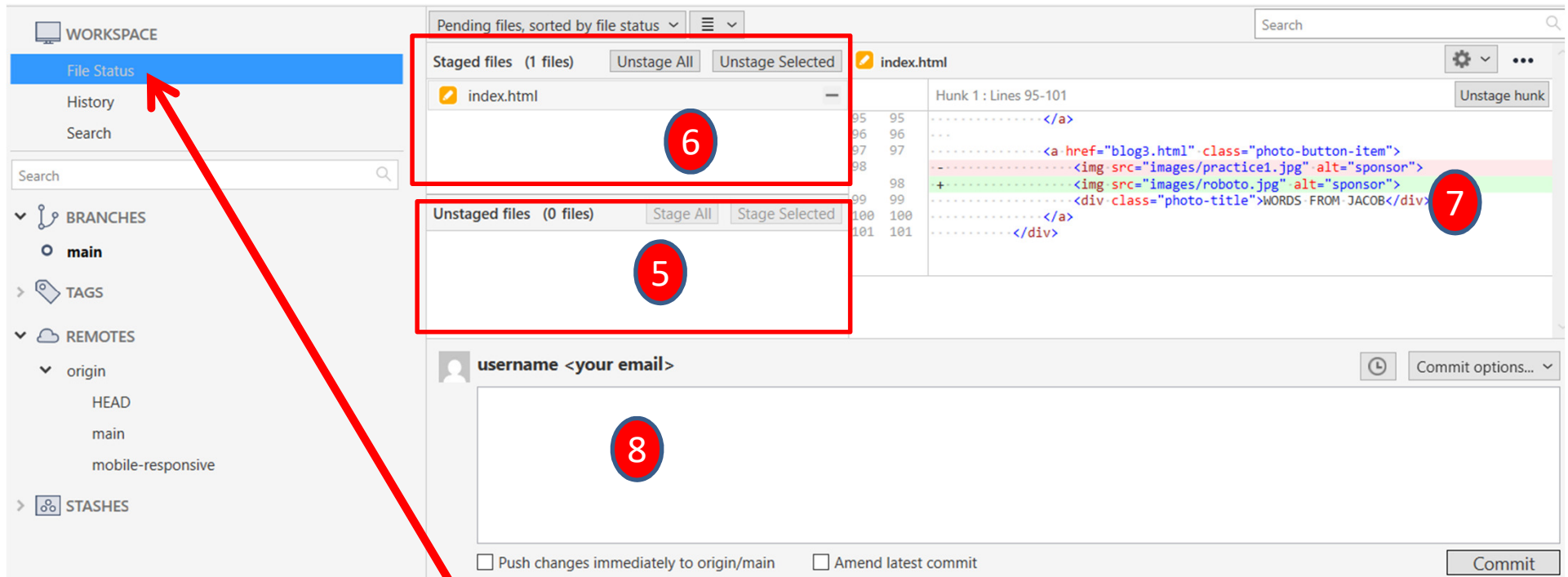
- A commit is a group of files saved to github at the same time
- Fetch
 - Will ensure the list of commits is up to date. If there are changes in github, a small number will appear beside the Pull button.
- Pull
 - Will “pull” files from github to your laptop
- Push
 - Will “push” committed files from your laptop to github
- Branch
 - Before making changes, create a new branch with a unique name (eg: feature/rateupdates). This ensures no files are merged into the main branch without proper review.

The screenshot shows the GitHub Desktop application interface. On the left, the 'WORKSPACE' sidebar has 'History' selected. The main area is divided into several sections:

- Commits (4):** A list of commits with columns for Graph, Description, Date, Author, and Commit. The most recent commit is 'Corrected html errors in about, blogs and index. Added g' by jacobdunn, dated 26 Dec 2025 19:47.
- Staged files (6):** A list of files ready to be committed. In this case, 'index.html' is the only file staged.
- Unstaged files (5):** A list of files that are different between the local computer and the GitHub branch. Currently, there are no unstaged files.
- Diff View (7):** A view showing the changes made to the selected file, 'index.html'. It highlights the differences between the current state and the last commit, showing additions in green and removals in red.

In the Workspace, select History

- 4 List of commits made for this repository with relevant comments, the person who made the commit, and the date of the commit
- 5 List of files that are different between your computer and the github branch
 - files can be staged (ready to commit) by clicking the **Stage Selected** button
- 6 List of files that are ready to commit
 - files can be removed from the commit by clicking the **Unstage Selected** button
- 7 Changes made to the selected file
 - red: code removed
 - green: code added

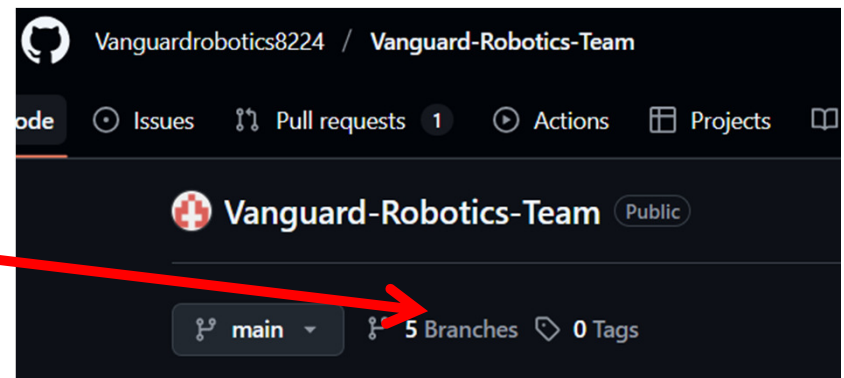


In the Workspace, select File Status

- 5 List of files that are different between your computer and the github branch
 - files can be staged (ready to commit) by clicking the **Stage Selected** button
- 6 List of files that are ready to commit
 - files can be removed from the commit by clicking the **Unstage Selected** button
- 7 Changes made to the selected file
 - red: code removed
 - green: code added
- 8 Type a relevant comment here, then press the **Commit** Button

Basic Procedure

1. Get the code: SourceTree
 - Select the working branch (usually main)
 - Click the **Fetch** button
 - If your laptop is out of date, click the **Pull** button to synchronize with github
 - Create a new branch (use the Branch button): call it your name
2. Start coding: VSCode!
 - See Karina's examples
3. Put changes in github: SourceTree
 - **Stage** files (check changes first!)
 - Write a short description of the changes
 - **Commit** the files
 - **Push** the commit to github
 - (you can verify they are there by going to your branch in github)

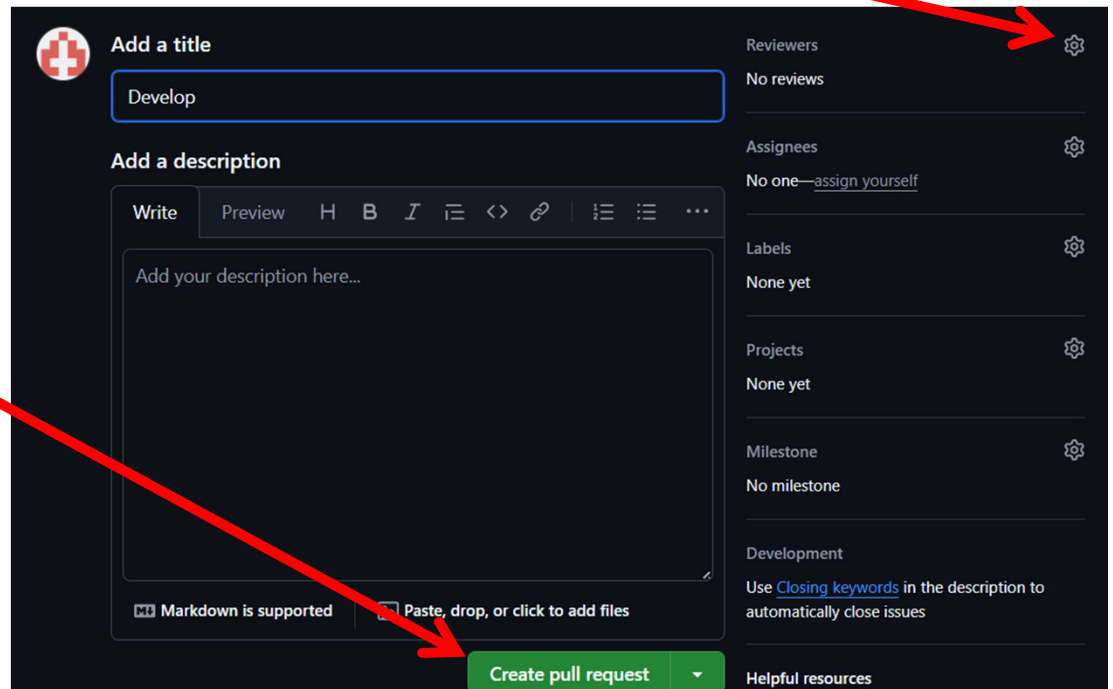
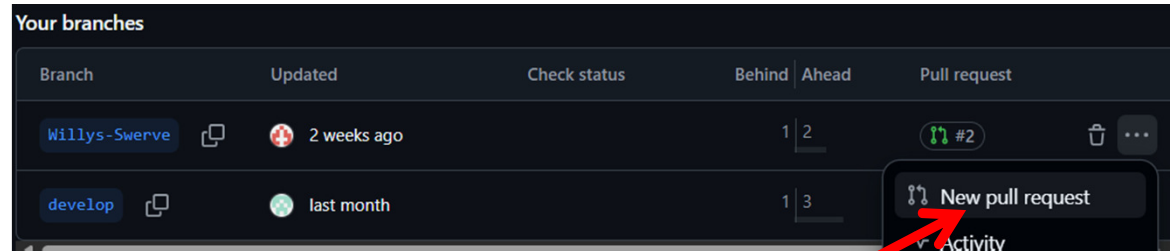


Ask for a code review

“correcting your homework”

- Select the new branch
- Click on the **three dots**, then **New Pull Request**
- Add a title and description
- Select reviewers to check your changes
- Click on the

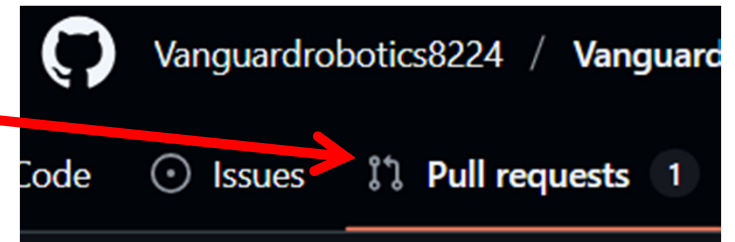
Create pull request
button



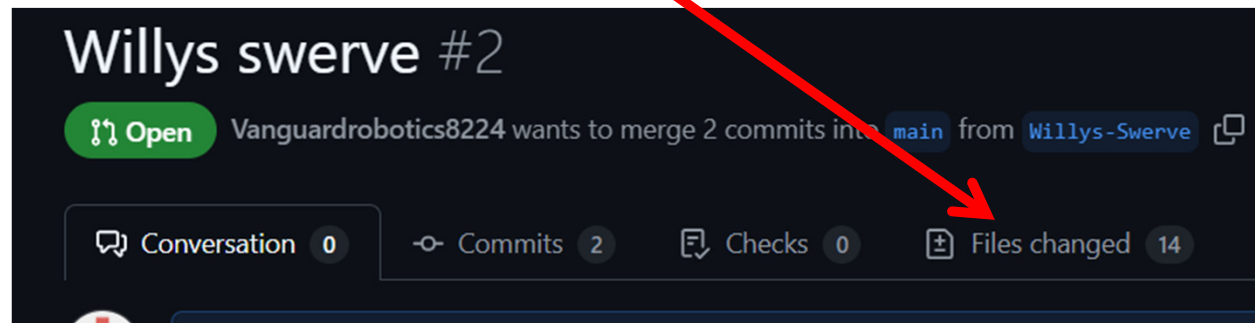
Reviewing a Pull Request

–do not review your own work

- Select **Pull requests**



- Select the pull request to review
- Select **Files changed**. A list of the new/deleted/changed files will appear



Reviewing a Pull Request

0 / 14 files viewed

Ask Copilot

Review in codespace

Review changes

Finish your review

Write

Preview

Leave a comment

Markdown is supported

Paste, drop, or click to add files

☒ **Comment**
Submit general feedback without explicit approval.

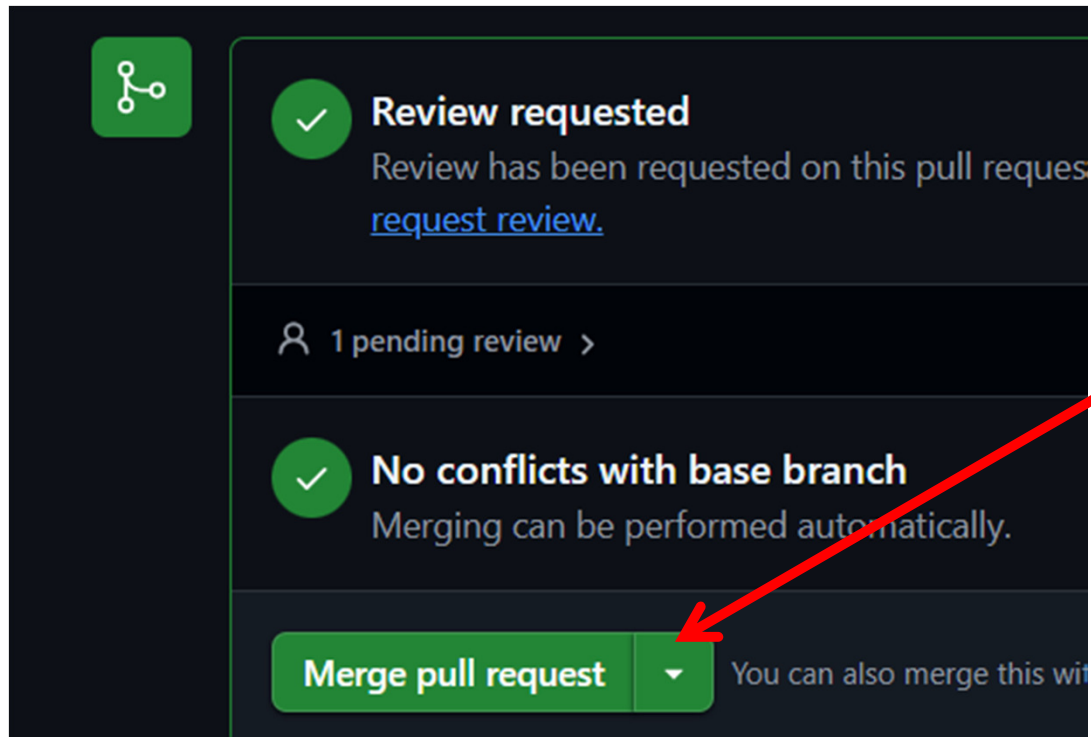
☐ **Approve**
Pull request authors can't approve their own pull request.

☐ **Request changes**
Pull request authors can't request changes on their own pull request.

Submit review

- Select **Review changes**
- There are three choices
 1. Leave a **comment**
 2. **Approve** the changes
 3. **Request changes**
(make the request in the comment box)
- Select **Submit review**

Merging a Pull Request



After your pull request has been approved by a reviewer:

- If there are conflicts, they must be resolved before you can merge
- If there are no conflicts, click **Merge pull request**
 - Your pull request will be merged into the main branch
 - Delete your branch

Congratulations!